

Dhruv Nirmal

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PROFESSIONAL SUMMARY

Data scientist who builds ML that moves a real business number. My clearest proof: a one-vs-all classifier (scikit-learn, TF-IDF to embeddings, GridSearchCV) that cut a client's **A\$12B** spend categorisation from a month to a single day's run, plus Prophet forecasting and pricing analysis that surfaced roughly **A\$2M** in savings. I work in Python and SQL on a Master of Data Science foundation, and I use AI agents daily, having built multi-agent QA and reporting systems now used company-wide, not just occasional AI-assisted coding.

KEY SKILLS

ML & Statistics: scikit-learn (TfidfVectorizer, LogisticRegression, GridSearchCV), classification, Prophet time-series forecasting, feature engineering, class-imbalance handling, cross-validation, anomaly detection, probability & statistics, linear algebra, NLP/embeddings

Python & Data: Python, pandas, numpy, SQL, relational databases, PySpark, Databricks, Snowflake

AI Tools & Agentic Systems: Claude, ChatGPT, multi-agent orchestration, agentic workflow design, prompt engineering, AI-output evaluation/QA, daily AI-assisted development

Cloud & Infrastructure: AWS (S3, Lambda, IAM, EC2), Azure, Docker, Terraform

Visualisation & Communication: Tableau, Power BI, data storytelling, stakeholder presentation

EXPERIENCE

Data Scientist, Purchasing Index Data Analytics (Comprara Group)

Jun 2024 to Present | Melbourne, Australia

A data and analytics consultancy. I own client engagements end to end: framing the problem, building the model, and presenting results to commercial stakeholders. Selected engagements:

ML spend classification: enterprise procurement client (~A\$12B spend)

- Problem: categorising the client's five-year, roughly A\$12B spend book was manual and took an account manager over a month per cycle, with only 60 to 65% reliably categorised.
- Approach: led the firm's first ML classifier, a one-vs-all model that keeps high-value spend on manual review and models the long tail automatically. scikit-learn, text features (TF-IDF moving to embeddings) plus engineered features, LogisticRegression with class-imbalance handling, tuned with GridSearchCV and cross-validation.
- Validation: evaluated on precision, recall and F1 rather than raw accuracy, given the heavy class imbalance across categories.
- Result: cut the categorisation cycle from over a month to a single day's model run, and documented the pipeline for reuse across the wider client base.

Demand forecasting: fresh-produce and agricultural client

- Problem: the client needed to plan raw-material and chemical inventory three months ahead, including chemicals sourced internationally.
- Approach: built a Prophet time-series forecasting model, incorporating external drivers such as sea-freight trends and input prices; prepared and engineered features with pandas, and delivered the output through a planning dashboard.
- Result: forecasts within a 12.5 to 14% error margin, comfortably inside the client's accepted tolerance.

Pricing and cost analytics: multi-venue restaurant group (AU/NZ)

- Problem: the client felt supply costs were too high but could not see where the money was leaking.
- Approach: owned the account solo, built a product catalogue of roughly 3,000 items, then a pricing-decision dashboard tracking price movement and cost leakage by product, venue and supplier (Python, SQL, statistical analysis); presented findings directly to the Chief Procurement Officer and category managers.
- Result: helped the client identify close to 30% savings, roughly A\$2M over a year, in their largest category.

AI tools and agentic workflows: internal, my own firm

- Use AI tools and agents daily as part of my own development workflow, not only as a productivity aid.
- Designed a five-agent QA validation workflow (Claude, Python) that checks client datasets for issues before delivery.
- Built the agentic report-generation workflow that turns analytics outputs into structured client progress reports; rolled out company-wide across the analyst team, saving roughly 12 to 15 hours a week across the firm.
- Also placed first in an internal AI hackathon with a call-transcript sentiment and quality analyzer, later adopted into the team's workflow.

Data pipelines and reporting: 5 enterprise clients

- Manage 20+ automated ETL pipelines (Python, SQL) processing millions of procurement transactions monthly, and maintain 16+ Tableau dashboards used daily by procurement and finance stakeholders.
- Applied statistical anomaly detection across procurement datasets, flagging cost-leakage patterns that contributed to roughly 15% savings for one client.

Data Engineer Intern, Victorian Centre for Data Insights (VCDI)

Aug 2023 to Nov 2023 | Melbourne, Australia

- Built a distributed anomaly-detection pipeline in Databricks using PySpark, engineering scalable transformation layers to process government procurement data, improving detection accuracy by about 20%.
- Delivered a Power BI analytics solution adopted by senior Department of Transport stakeholders, presenting model outcomes and system architecture to a cross-functional, non-technical audience.

Research Data Analyst, Terminal Ballistics Research Laboratory (TBRL), DRDO

Jan 2021 to Jul 2021

- Applied statistical modelling and signal filtering (Butterworth filter) to blast-wave data to predict noise levels, improving prediction accuracy by about 20%, and compiled findings into a 70-page technical report.

PROJECTS

Job Hunt OS, agentic AI career-search system (Claude Code, multi-agent orchestration, Python)

- Designed and built a multi-agent system, an orchestrator plus specialist agents for resume tailoring, outreach and career coaching, that runs natively on Claude Code with no paid APIs, tailors resumes and cover letters to a job description, and produces ATS-ready two-page PDFs end to end.

Facit, profitability analytics SaaS for independent cafes (Python, forecasting, BI)

- Built and own a multi-tenant product that unifies POS transactions, staff wages and 14 suppliers' invoices into one weekly profitability view (revenue, wage cost %, food cost %, gross margin), with revenue forecasting, menu engineering and supplier price-creep detection.
- Live in day-to-day use with a design-partner cafe (Neighbours Cafe, St Kilda); I define the metrics it tracks and make the product decisions myself, weekly, based on real usage.

EDUCATION

Master of Data Science, Monash University | Feb 2022 to Dec 2023

Coursework: Statistics I & II, Machine Learning, High-Dimensional Data Analysis, Applied Forecasting, Communicating with Data.

Bachelor of Engineering, Thapar University | May 2017 to May 2021